Deadline : Wednesday, 24 June 2026 (10:00 AM)
Submission Process 1. Create a public GitHub repository for this assignment 2. Add a **proper and clear** ReadMe.md 3. Submit the GitHub repository link and other required details in the provided Google Form here 4. Please fill up the Google Form using the email you used in your initial application. 5. You can submit your response only once.
***Contact only in case of emergencies or issues with the above email here

Overview

This assignment includes one key task to assess your skills as a Software Developer.

- Ensure your project is functional and can be run on our end without issues.
- Hard coding is not allowed. Use dynamic, configurable, or parameterized approaches.
- If you have any questions or concerns about this assignment, you can email us. However, ensure you have thoroughly read the instructions before reaching out.
- For the frontend, use **React.js** for UI components. If you are implementing a backend, use **Django (optional)**.
- Even if you are unable to complete the whole assignment, you can still submit it.

Good Luck!

Task

The **dataset** for this task can be accessed here

You are provided with a dataset containing employee shift records across multiple dates. Each record includes a date, shift start date/time, shift end date/time, duration, and an activity reason such as Breakdown, Power Failure, Other, and more.

Your task is to build a web application that enables users to visualize shift patterns, analyze operational performance, and generate actionable insights through filtering, aggregation, and streak analysis.

- The **dataset may contain operational inconsistencies**. **Detect, document, and handle them appropriately.**
- Users should be able to **filter and analyze data** using combinations of relevant attributes.
- Provide **visualizations** that help users understand operational activity over time.
  - A shift analysis chart (**see the figure below for an example**) based on the clean **consistent data.**
  - Add more visualizations if you have more ideas.
- The operations team has observed **recurring breakdown periods**. Design and implement a method to **identify meaningful breakdown streaks** in the system and display them accordingly. Clearly document your assumptions and reasoning.
- Calculate and Display
  - '**Operational Efficiency Score = (Productive Hours / Total Hours) × 100**' where, **Productive Hours** = Hours where activity is not Breakdown or Unknown Failure and **Total Hours** = Sum of all shift hours
- Provide at least three operational insights that a plant manager could act upon.
- Design a solution that remains useful when new activity categories appear or categories are grouped together.

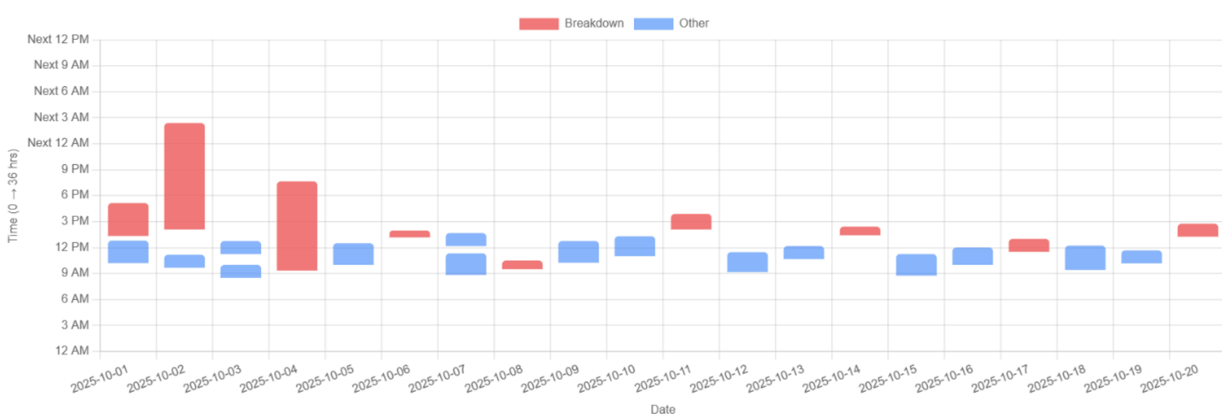

Figure: Example of a shift analysis chart **(two reasons are shown here for example, but the visualization must include all reasons in the dataset).**